

AI Usage DISCLOSURE FORM

1. Team Details

Team Name: Thapar_Bandits

Project / Product Name: Agentic Code Intelligence

Organization / Institution (if any): Thapar Institute Of Engineering And Technology

Submission Date: 30-09-26

2. AI Usage Declaration

Did your team use any Artificial Intelligence (AI) in developing this project? Yes / No

Yes

3. Purpose of AI Usage (Brief Details)

Idea generation / brainstorming : Used for structuring RAG pipeline & evaluating retrieval benchmarks.

Code generation or assistance : Assisted in writing MTEB encoder wrapper script & CLI search demo.

UI / UX design : N/A

Content creation : N/A

Data analysis : Analysis of NDCG@10 and MRR@10 evaluation metrics from CoIR dataset.

Testing / debugging : Benchmarking retrieval performance via MTEB library.

Other : N/A

4. Feature Origin Classification (Please add/delete based on the number of the feature)

Feature 1

1. Feature Name: Dual Pipeline Query & Code Preprocessing Engine

2. Self-Generated/AI-Generated/Both : Both

3. Description (include AI tools/platform Used, prompt used , Output Summary, Modification) :

- **AI Tools / Platform Used:** ChatGPT / Gemini / GitHub Copilot
- **Prompt Used:** *"How to separate query preprocessing from code preprocessing and implement camelCase identifier splitting in Python?"*
- **Output Summary:** Helper functions for intent normalization and code tokenization (USE_IDENTIFIER_SPLIT).
- **Modification:** Customized regex rules to handle variable names without breaking math expressions.

Feature 2

1. Feature Name: Overlapping Sliding-Window Chunking (30-line window, 15-line stride)

2. Self-Generated / AI-Generated / Both: Self-Generated

3. Description:

- **AI Tools / Platform Used:** None (Built manually)
- **Prompt Used:** N/A
- **Output Summary:** Python file-reading engine that cuts source code files into 30-line overlapping windows.
- **Modification:** Logic created natively to ensure split functions stay intact across chunk boundaries.

Feature 3

1. Feature Name: Vector Encoding & Cosine Similarity Search (BAAI/bge-small-en-v1.5 + NumPy)

2. Self-Generated / AI-Generated / Both: Both

3. Description:

- **AI Tools / Platform Used:** Hugging Face , sentence-transformers , mteb
- **Prompt Used:** *"Write an MTEB AbsEncoder class wrapper for sentence-transformers on CPU."*
- **Output Summary:** Encoder wrapper class PrePostPipelineEncoder for benchmark evaluation.
- **Modification:** Configured NumPy normalized dot-product search to run strictly on CPU without GPU overhead.

5. Ethical & Compliance Confirmation

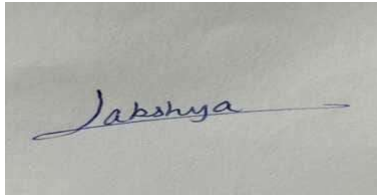
AI usage complies with guidelines and policies. **Yes**

No proprietary or copyrighted data misused. **I Agree**

6. Declaration & Sign-Off

Name of Team Representative: LAKSHYA GARG

Role: TEAM LEAD

A photograph of a handwritten signature in blue ink on a light-colored surface. The signature is cursive and appears to read 'Lakshya'.

Signature:

Date:30-09-26